Replication and Enhancement of All Weather Portfolio

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Background & Rationale

Ray Dalio developed the **All Weather Portfolio** to solve a fundamental family problem: how to invest reliably without constant market timing or guidance.

- **Passive Management:** Designed for investors who cannot (or should not) time the market.
- **Purchasing Power:** Protection against high-inflation periods where cash fails.
- **Outperform 60/40:** To have less risk than traidtional mix while matching the returns

"I believe that for most investors managing their own portfolios, investing should be done with little or no market timing." — Ray Dalio

Designing for All "Seasons"

Rising Growth Equities, Commodities, Corporate Credit	**Rising Inflation** Gold, Commodities, ILBs
Falling Growth Treasury Bonds, ILBs	**Falling Inflation** Stocks, Nominal Bonds

Mechanics of Risk Parity

Balancing Volatility

Risk parity means taking investments of different volatilities and normalizing them so they contribute equal risk to the portfolio.

Engineering Levels

We "level up" low-risk/low-volatility assets (like bonds) and "level down" high-risk assets (like stocks) so their impacts are balanced.

Fundamental Drivers

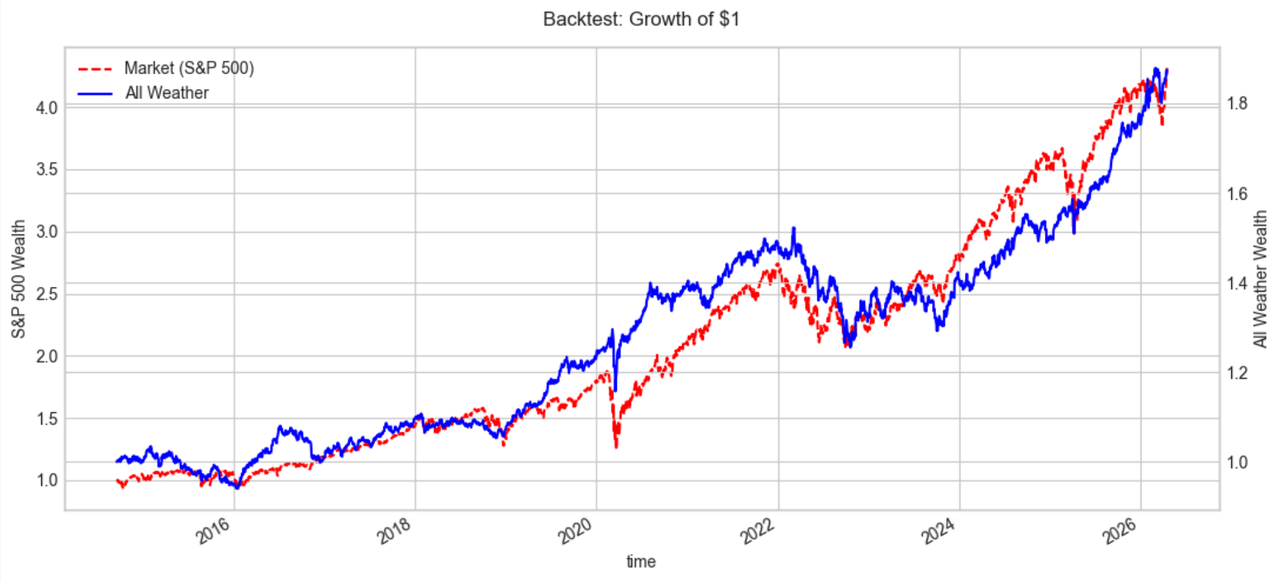
Exposures are balanced based on how each asset class responds to fundamental changes in **Inflation** and **Growth**.

$$\text{Asset Risk Contribution} = \text{Weight}_i \times \text{Volatility}_i \rightarrow \text{Constant across all classes}$$

Strategies Comparison

Strategy	Pros	Cons
All Weather	Passive, low volatility	Underperform when equity runs
Buy and Hold	Passive, Beta chasing	Capital lockup, market crash
Discretionary Macro	Liquidity, profit from dislocations	Fail in QE, cross domain knowledge demanded
Trend Following	Simple, behavioral	Regime change, turnover
Statistical Arbitrage	Extreme low market Beta	Maths, High Transaciton Costs
Equity Long Short	Low volatility, low Beta	Fundamental Research

Replication



	Full Hist. Annual SD (%)		Portfolio Weight (%)		Assigned Vol (%)
SPY – stocks	17.63%		14.36%		2.53%
GLD – gold	15.91%		15.91%		2.53%
DBC – commodities	17.83%	X	14.19%	=	2.53%
IEF – intermediate bonds	6.57%		38.52%		2.53%
TLT – long term bonds	14.86%		17.02%		2.53%

Risk Parity Allocation

1. Compute annualized volatility

$$\hat{\sigma}_i = s_i\sqrt{252}$$

2. Assign weight proportional to inverse volatility, So that riskier assets receive smaller weights

$$w_i^* = \frac{1}{\hat{\sigma}_i}$$

3. Normalize the raw weights so they sum to 100%

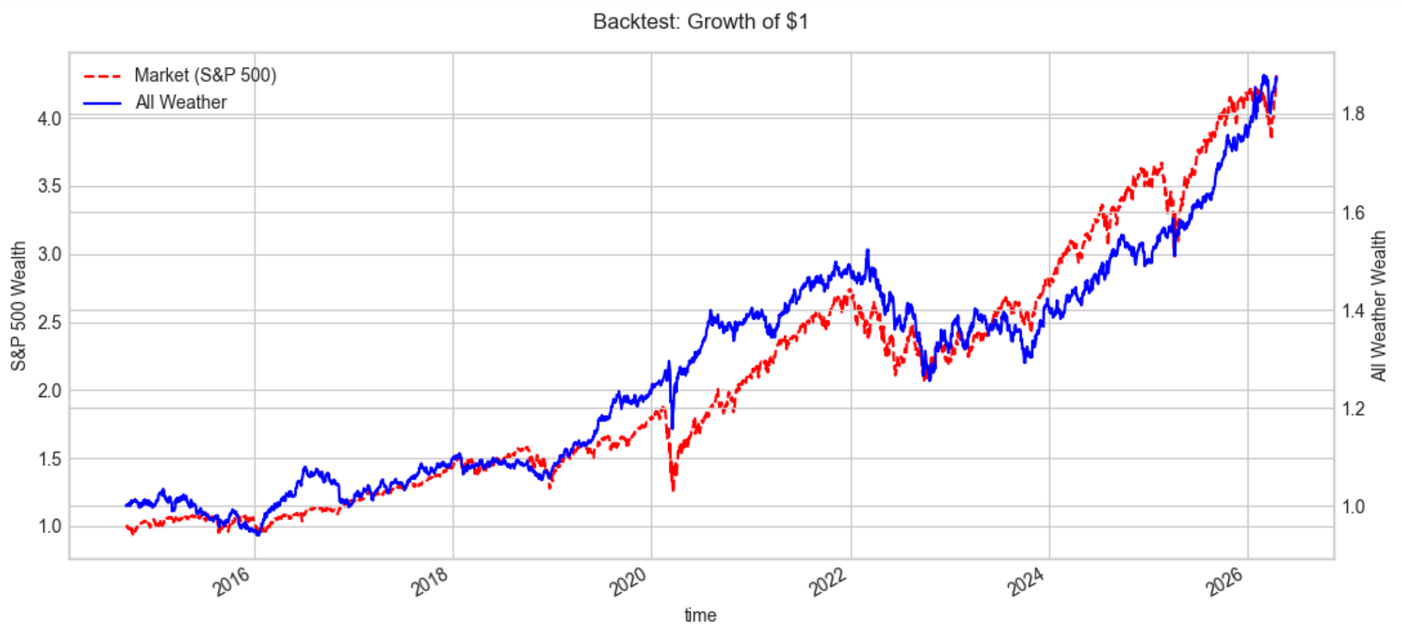
$$w_i = \frac{w_i^*}{\sum_{j=1}^n w_j^*} = \frac{1/\hat{\sigma}_i}{\sum_{j=1}^n 1/\hat{\sigma}_j} \rightarrow \sum_{i=1}^n w_i = 1$$

4. Each assigned ~2.5% volatility

Performance

Summary Statistics

	Total Return (%)	CAGR (%)	Vol (%)	SR (raw)	SR (rf=4%)	MDD (%)
Market (S&P 500)	331.52%	13.49%	17.63%	0.77	0.54	-33.72%
AWP	87.42%	5.59%	7.15%	0.78	0.22	-17.56%



AWP vs Market?

- Lower return / lower risk
- Slightly outperformed at risk-adjusted basis

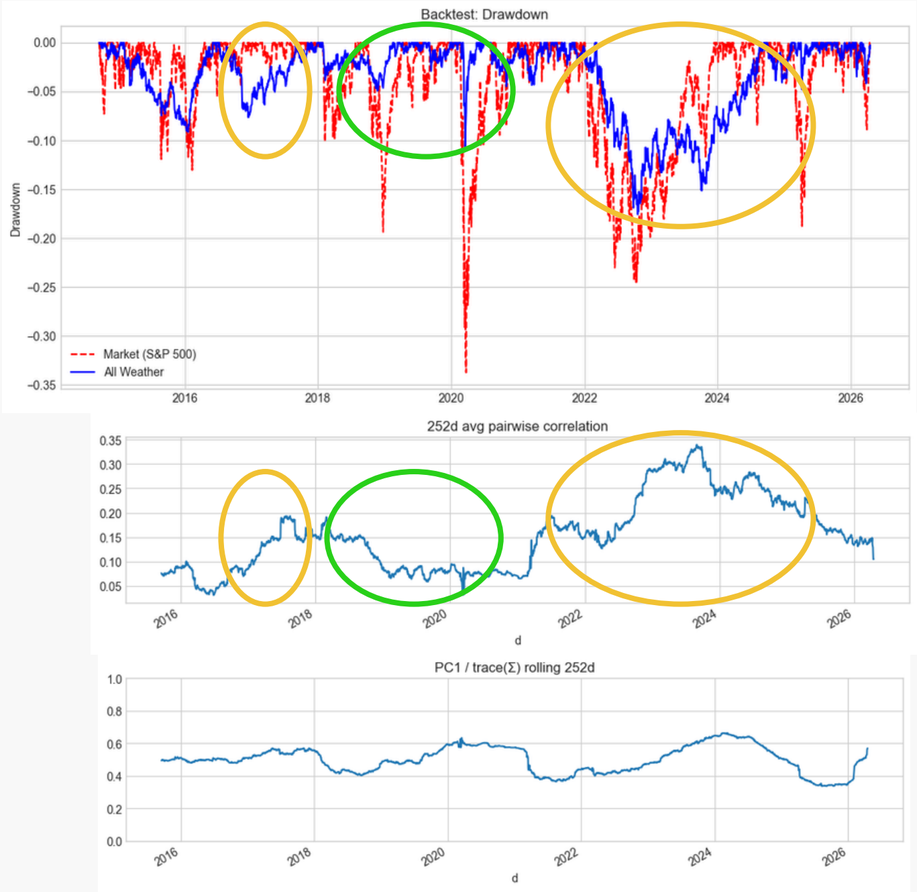
Load it up?

- Assumed 4% risk-free rate (borrowing cost)
- AWP underperformed by more than half
- 2.46x to get the same level of vol
- +2.32% increase only in expected return

- 7.91% vs 13.49%

All weather =
Premium paying for protection?

Protection



Drawdown protection?

- Occationally
- Recent years worse than market

Diversification?

- Protection is gone when correlation spikes
- >40% is exposed to the same factor (PC1)

- May not worth for the return given up?

Factor Loadings

```
==============================================================================
Dep. Variable:                      Y   R-squared:                       0.138
Model:                            OLS   Adj. R-squared:                  0.136
Method:                 Least Squares   F-statistic:                     53.45
Date:                Fri, 24 Apr 2026   Prob (F-statistic):           2.52e-61
Time:                        03:38:43   Log-Likelihood:                 8315.1
No. Observations:                2003   AIC:                        -1.662e+04
Df Residuals:                    1996   BIC:                        -1.658e+04
Df Model:                           6
Covariance Type:            nonrobust
==============================================================================
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
const       6.733e-05   8.54e-05      0.788      0.431      -0.000       0.000
Mkt-RF         0.1221      0.008     15.888      0.000       0.107       0.137
SMB            0.0080      0.014      0.553      0.580      -0.020       0.036
HML           -0.1205      0.014     -8.494      0.000      -0.148      -0.093
RMW           -0.0171      0.019     -0.891      0.373      -0.055       0.021
CMA            0.2168      0.026      8.184      0.000       0.165       0.269
Mom           -0.0078      0.009     -0.880      0.379      -0.025       0.010
```

- Slight loading to equities (0.12)
- HML (-0.12)
- CMA (0.22)

* No alpha either

In general, not explained well by equity factors (14% R2)

"Seasonal" Rotations

All Weather Thesis

	Inflation	Growth
Equties	Neg	Pos
Treasury Bonds	Neg	Neg
Gold	Pos	.
Commodities	Pos	Pos

Empirical Loadings

	Inflation	Growth
SPY – stocks	-0.18	-0.03
TLT – long term bonds	** -0.43	0.06
IEF – intermediate bonds	** -0.48	0.04
GLD – gold	* -0.30	0.04
DBC – commodities	*** 0.71	0.03
AWP	-0.67	0.14

- Equities, Treasury, Commodities act as intended
- Gold is not a protection for inflation
- No individual asset is sensitive to growth
- AWP is short inflation, long growth

Possible Advancement

Adding Crypto

- Delivers exponential returns while reducing weights to balance the risks

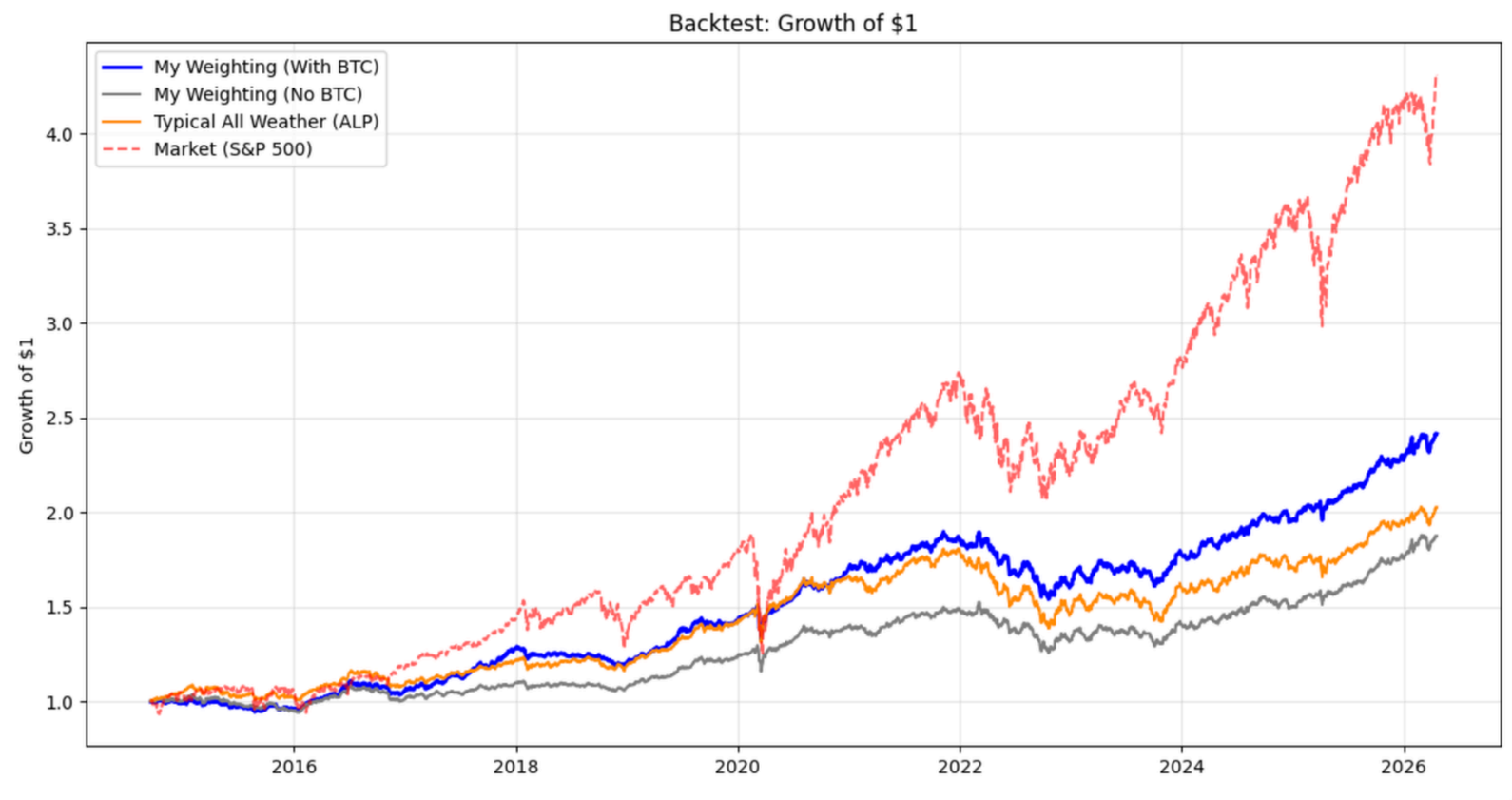
Adding SVXY

- Volatility tends to go down by time

Rebalancing frequently

- Prevents single-asset risk concentration for the asset class gaining sudden surges in between rebalancing periods

Crypto



	Full Hist. Annual SD (%)	Portfolio Weight (%)
SPY - stocks	17.62	13.83
GLD - gold	15.91	15.32
DBC - commodities	17.83	13.67
IEF - intermediate bonds	6.57	37.11
TLT - long term bonds	14.86	16.40
btc	66.50	3.67

- BTC as a new asset class delivers potentially outstanding return
- Balance between risk and return
- Each asset class accounts for ~2.4% volatility

Crypto

Summary Statistics

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================================================================================
BTC PERFORMANCE SUMMARY
================================================================================
                 Strategy  Total Return (%)  CAGR (%)  Vol (%)  Sharpe  MDD (%)
        Market (S&P 500)            331.52     13.46    17.63    0.52   -33.72
     Typical AWP (Dalio)            102.63      6.29     8.51    0.24   -23.37
  Our Weighting (No BTC)             87.42      5.58     7.15    0.18   -17.56
Our Weighting (With BTC)            141.71      7.92     7.59    0.48   -18.95
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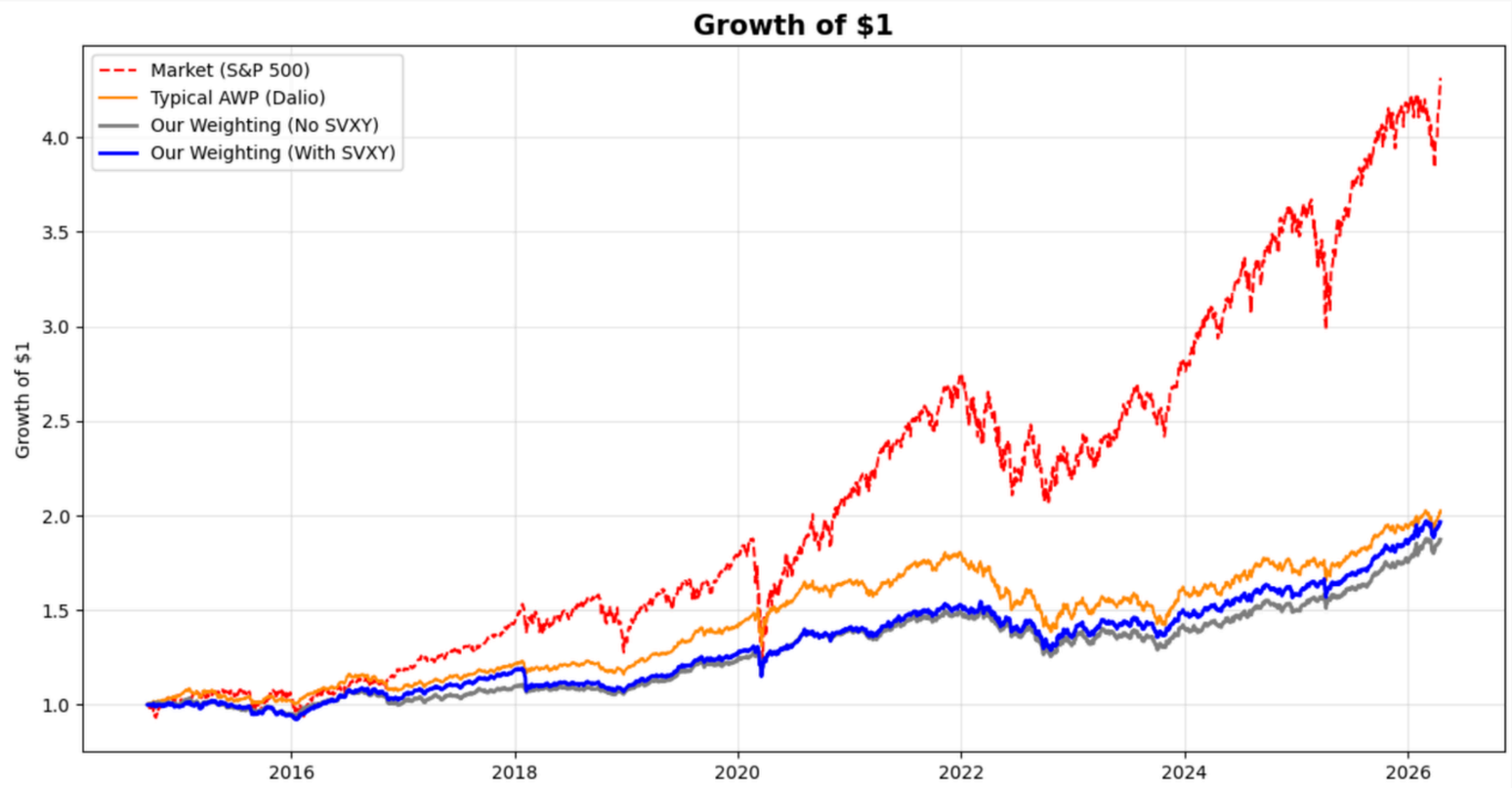
Potential Limitations

- Skeptism on Crypto → Trump's 2th supports crypto
- No holding returns, only capital gains → Institutions launches ETFs with dividends

SVXY



	Full Hist. Annual SD (%)	Portfolio Weight (%)
SPY - stocks	17.63	13.72
GLD - gold	15.91	15.20
DBC - commodities	17.83	13.56
IEF - intermediate bonds	6.57	36.81
TLT - long term bonds	14.86	16.27
SVXY	54.50	4.44

- Volatility tends to go down in long run
- Long SVXY allows for
  - Avoiding short interests
  - Eliminates unlimited downside
- Each asset class accounts for ~2.4% volatility

SVXY

Summary Statistics

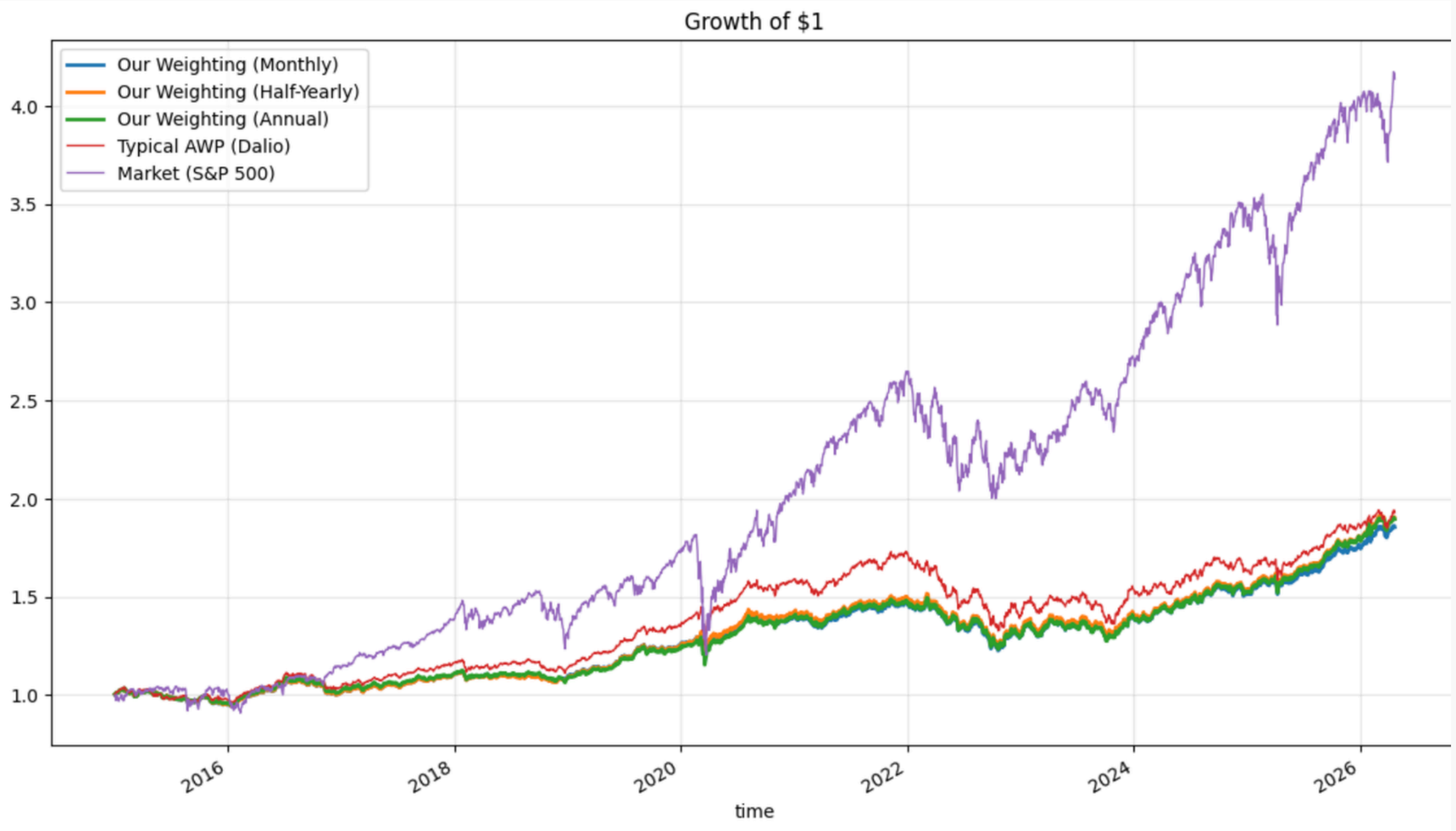
SVXY PERFORMANCE SUMMARY

Strategy	Total Return (%)	CAGR (%)	Vol (%)	Sharpe	MDD (%)
Market (S&P 500)	331.52	13.46	17.63	0.52	-33.72
Typical AWP (Dalio)	102.63	6.29	8.51	0.24	-23.37
Our Weighting (No SVXY)	87.42	5.58	7.15	0.18	-17.56
Our Weighting (With SVXY)	96.67	6.02	7.65	0.23	-16.71

Potential Limitations

- Sudden VIX surge in Black Swan Events → Long SVXY only after VIX reaches certian level
- No holding returns, only capital gains → Short option premiums

Frequent Rebalancing



- Reduce assets risk exposure to sudden momentum inter-period
- E.g 2025 Q4 Gold; 2026 Mar Oil
- Flexibility adjustment in response to macro sudden changes
- Limitations:
  - Rebalancing costs

Takeaways

Portfolio Performance Backtest Summary (2015-2026)

Strategy	Total Return (%)	CAGR (%)	Vol (%)	Sharpe	MDD (%)
All (Annual)	170.07	9.19	8.24	0.6	-19.3
All (Half-Yearly)	165.72	9.03	8.11	0.59	-19.26
All (Monthly)	160.48	8.84	8.09	0.56	-19.07
Core + BTC (Annual)	158.37	8.76	7.68	0.58	-19.54
Core + BTC (Half-Yearly)	151.81	8.52	7.59	0.56	-19.56
Core + BTC (Monthly)	144.79	8.25	7.56	0.52	-19.4
Market (S&P 500)	313.64	13.39	17.71	0.51	-33.72
Core + SVXY (Half-Yearly)	104.36	6.53	7.68	0.29	-17.09
Core + SVXY (Annual)	102.04	6.42	7.79	0.27	-17.05
Core + SVXY (Monthly)	101.27	6.39	7.65	0.28	-16.86
Typical AWP (Dalio)	92.85	5.98	8.57	0.2	-23.37
Core (Half-Yearly)	89.87	5.84	7.17	0.22	-17.9
Core (Annual)	89.42	5.82	7.24	0.21	-17.58
Core (Monthly)	85.39	5.62	7.12	0.19	-17.89

Improvements evidenced by higher SR

Thank you